

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

COURSE CODE: CSA1407

COURSE NAME: COMPILER DESIGN

### COURSE OUTCOME COVERED

CO3: Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks:100

Annexure A

### EXPERIMENTS

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#### Code of Conduct:

I *M.V.Rakesh Reddy* (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:***M.V.Rakesh Reddy*

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## Annexure A

### EXPERIMENTS

Q1. You are developing a compiler-based desk calculator that evaluates arithmetic expressions using syntax-directed definitions (SDD).

The system should:

- (i) Accept an arithmetic expression with operators +, -, \*, / and parentheses
- (ii) Use syntax-directed evaluation to compute values based on operator precedence
- (iii) Optimize evaluation by reducing unnecessary temporary computations
- (iv) Display the final result of the expression

Demonstrate how SDD rules (synthesized attributes) are used for expression evaluation.

SDD

Grammar with semantic rules:

$E \rightarrow E+T \quad \{E.val = E1.val + T.val \}$

$E \rightarrow E-T \quad \{E.val = E1.val - T.val \}$

$E \rightarrow T \quad \{E.val = T.val \}$

$T \rightarrow T * F \quad \{T.val = T1.val * F.val \}$

$T \rightarrow T / F \quad \{T.val = T1.val / F.val \}$

$T \rightarrow F \quad \{T.val = F.val \}$

$F \rightarrow (E) \quad \{F.val = E.val \}$

$F \rightarrow \text{digit} \quad \{F.val = \text{digit} \}$

Q2. In a compiler, postfix expressions are evaluated during code generation using stack-based bottom-up evaluation.

Write a C program that:

- (i) Evaluates a postfix expression
- (ii) Demonstrates S-attributed definition evaluation
- (iii) Shows how intermediate results are computed

Q3.You are designing a parser that uses L-attributed definitions, where inherited attributes are passed from parent to child.

Write a C program that:

- (i).Evaluates a simple expression like  $a + b * c$
- (ii) Uses function parameters to simulate inherited attributes
- (iii). Ensures correct precedence handling
- (iv). Demonstrates top-down evaluation logic

Explain how inherited attributes influence computation.

Q4.In a compiler, it is important to check whether two type expressions are equivalent.

Write a C program that:

- (i).Accepts two type expressions (e.g., int, float, int\*)
- (ii).Compares them for **type equivalence**
- (iii).Displays whether they are:

(a).Equivalent

(b).Not equivalent

Extend the logic to handle pointer types (int\*, float\*).

Q5.You are building a semantic analyzer that detects type errors in arithmetic expressions.

Write a C program that:

- (i). Accepts two operands and their types
- (ii). Checks whether the operation (e.g., +, \*) is valid
- (iii). Reports:
  - (a).Valid expression

(b).Type error (e.g., int + char\*)

Demonstrate how compilers detect semantic errors during type checking

## RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                              |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25        | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30        | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20        | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |
| Analysis of Results       | 15        | Insightful analysis with accurate interpretation and validation of results                    | Good analysis with reasonable interpretation                | Basic analysis with limited insights                | Poor or incorrect analysis of results            |
| Documentation             | 10        | Clear, well-structured documentation with diagrams, code, and results                         | Organized documentation with necessary details              | Basic documentation with missing details            | Poor or incomplete documentation                 |

Compiler Design Assess... CSA1407-Compiler... ACrOgDfd1S2XBpmyrl... CO3-AT1-Experiments - C... ACrOgAoasG1A7bgel... CSA14 - Compiler Design... Home - Canva

github.com/Rakeshreddymeruva/CSA1407-Compiler\_Design/tree/main/ATs/CO3\_AT1

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| Name       | Last commit message  | Last commit date |
|------------|----------------------|------------------|
| ..         |                      |                  |
| .gitkeep   | Create .gitkeep      | 1 minute ago     |
| SDD.c      | Add files via upload | now              |
| Lc         | Add files via upload | now              |
| post.c     | Add files via upload | now              |
| semantic.c | Add files via upload | now              |
| type.c     | Add files via upload | now              |

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**CSA1407-...**



1 Contributor 0 Issues 0 Stars 0 Forks

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